

Futureproof

Equity Preservation Mortgage v14c

Independent Actuarial Review

50,000-Path Monte Carlo with Sensitivity & Stress Analysis

April 2025

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Executive Summary

This report presents an independent actuarial review of the FutureProof Equity Preservation Mortgage (EPM) v14c (003) model. The model uses a **GBM + Mean Reversion** equity return model based on MLE parameter estimation (Shevchenko, April 2026), a 2.0% wholesale funding margin, and a $\pm 40\%$ –20% volatility buffer collar.

The conclusions in this report are conditional on the model parameter set. A separate companion paper (*Model Assumptions & Parameter Risk*, April 2025) performs independent empirical calibration and re-runs the headline PoD across the data-supported range for each input. That paper finds a materially wider plausible PoD range than is evident from the base case alone. Summary:

Parameter scenario	PoD (30yr)	Basis
Base case (model as written)	1.2%	Parameters per v14c-003
Realistic central	16.9%	Each input at empirical midpoint
Adverse plausible	43.2%	Each input inside empirical range, non-tail

This range is driven primarily by μ (equity drift), γ (mean-reversion strength), and collar cost. Readers should anchor capital and reinsurance sizing on the adverse-plausible leg; the base case is retained here for traceability against the production model, not as a central expectation.

Key Findings

- **Base case PoD: 1.2%** (SE: 0.05%) — headline from the model-as-written
- **Empirically-centred PoD: 16.9%** — each parameter at data-supported midpoint, joint re-run
- **Adverse plausible PoD: 43.2%** — each parameter shifted inside (not beyond) empirical range
- **Median surplus at maturity (base): \$2,151,975** — typical mortgage still generates substantial surplus
- **LMI fair premium (base): \$1,716** — commercially viable at base parameters; would rise materially under realistic-central
- **Primary uncertainty drivers:** equity drift μ , mean-reversion γ , and collar cost — each individually capable of shifting PoD by >5pp
- **Cash-rate θ is secondary but material:** +130bp lifts PoD from 1.2% to 6.4%
- **Correlation ρ , volatility σ , and κ are well-calibrated** and do not drive material uncertainty

Model Overview — v14c (003) Parameters

The table below summarises all key model parameters.

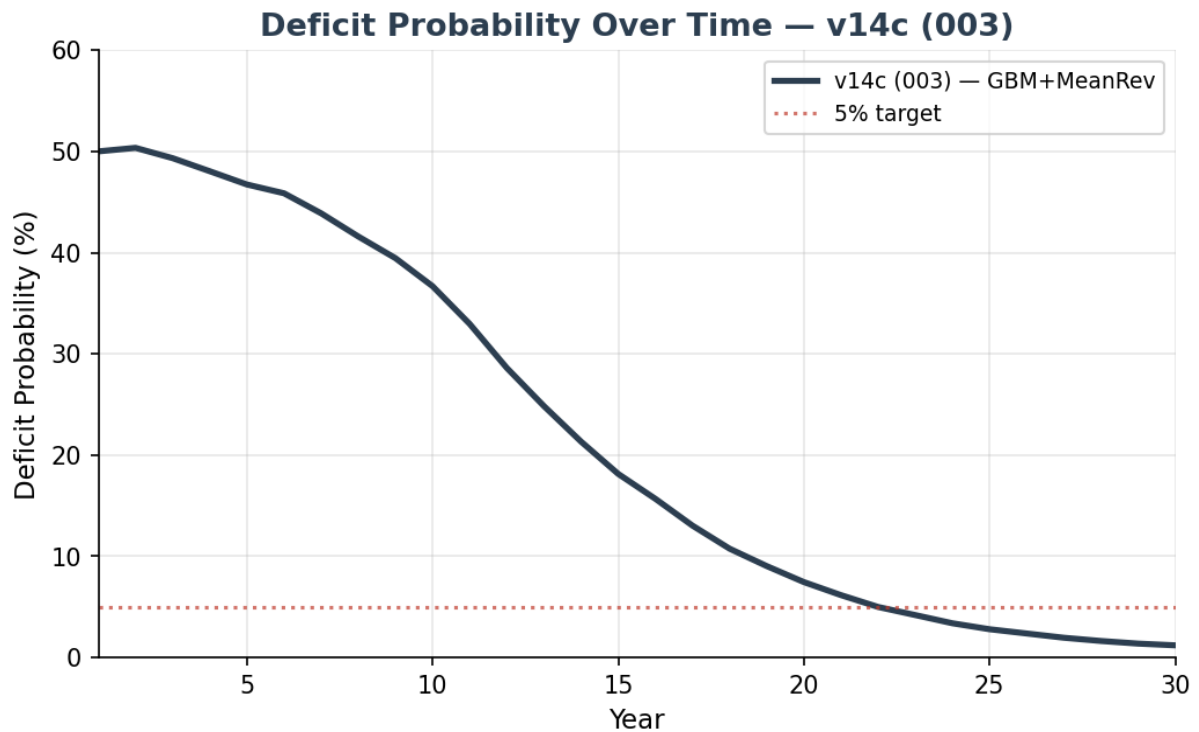
Parameter	Value
Home Value	\$2,000,000
LVR	80%
Initial Loan	\$1,300,000
Annuity	\$30,000/yr × 10yr
Equity Model	GBM + Mean Reversion (Shevchenko 2026)
Expected Return	9.2%
Equity Volatility	16.6%
Mean Reversion Speed (γ)	0.163
Buffer Cap / Floor	140% / 80%
Wholesale Margin	2.0%
NIM	0.70%
FP Margin	0.50%
Hedging Fee	0.25%
Total Variable Costs	3.45%
Cash Rate ($\theta / \kappa / \sigma$)	2.13% / 0.24 / 1.22%
Cash Rate Initial	4.21%
Correlation (equity-rate)	0.30
Collar Price	+0.046% (near zero net cost)
Simulation Paths	50,000

Note on the Mean-Reverting Equity Model: The Shevchenko (2026) paper proposes a novel model: $S(t+1) = S(t)(1 + \mu + \sigma\epsilon) + \gamma(M(t) - S(t))$, where $M(t)$ is a deterministic trend growing at the expected return rate. The mean reversion parameter $\gamma = 0.163$ means that roughly 16% of any deviation from trend is corrected each year. Pure GBM ($\gamma = 0$) lets deviations compound indefinitely.

Parameter-risk caveat: independent MLE on S&P 500 TR 1988–2024 gives $\gamma = 0.173$, which appears to confirm the model input. However, the 95% bootstrap CI runs [0.019, 0.447], and a likelihood-ratio test against $\gamma = 0$ yields $p = 0.09$ — not significant at the 5% level. A small-sample bias correction suggests the true γ most consistent with the data may sit closer to 0.05–0.10. At $\gamma = 0.10$, base-case PoD rises from 1.2% to 3.1%; at $\gamma = 0$, to 31.5%. γ is the single most load-bearing modelling choice in the product.

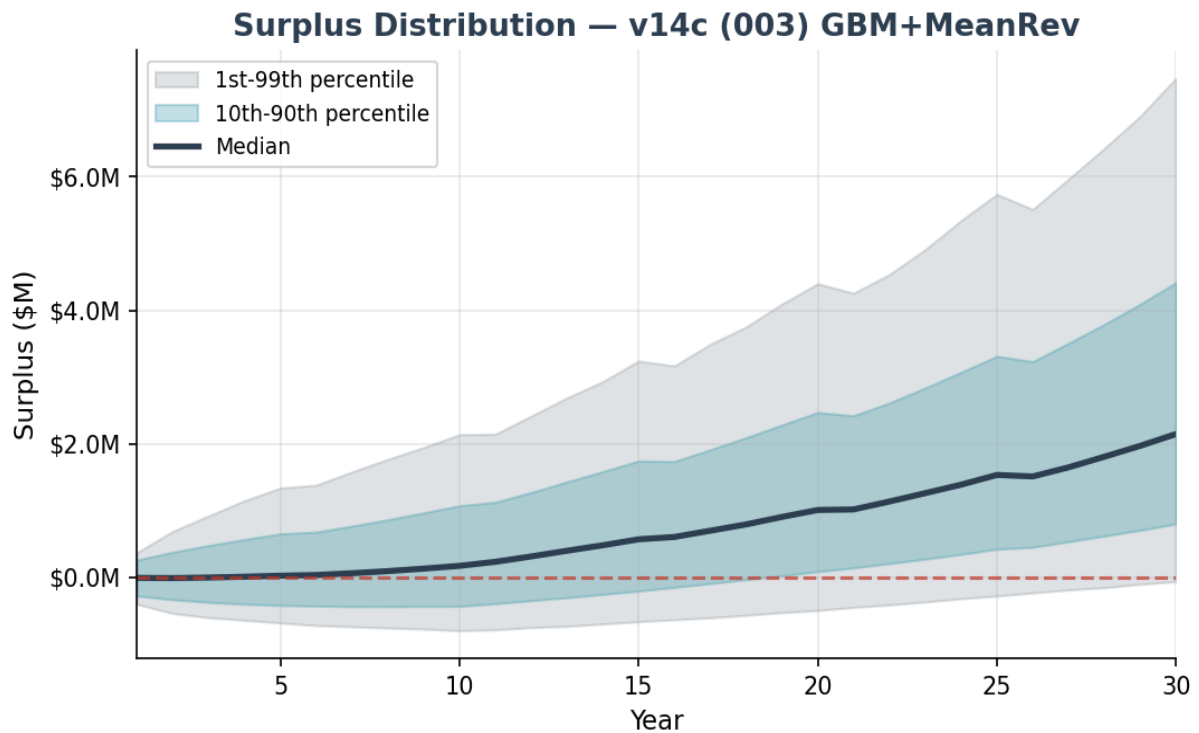
Base Case Results — 50,000 Paths

Deficit Probability Over Time



PoD starts at 50.01% in year 1 (reflecting upfront costs and annuity payments) and declines steadily to 1.2% by year 30. The steepest improvement occurs between years 10-20 as P&I amortisation reduces the mortgage balance and investment compounding accelerates.

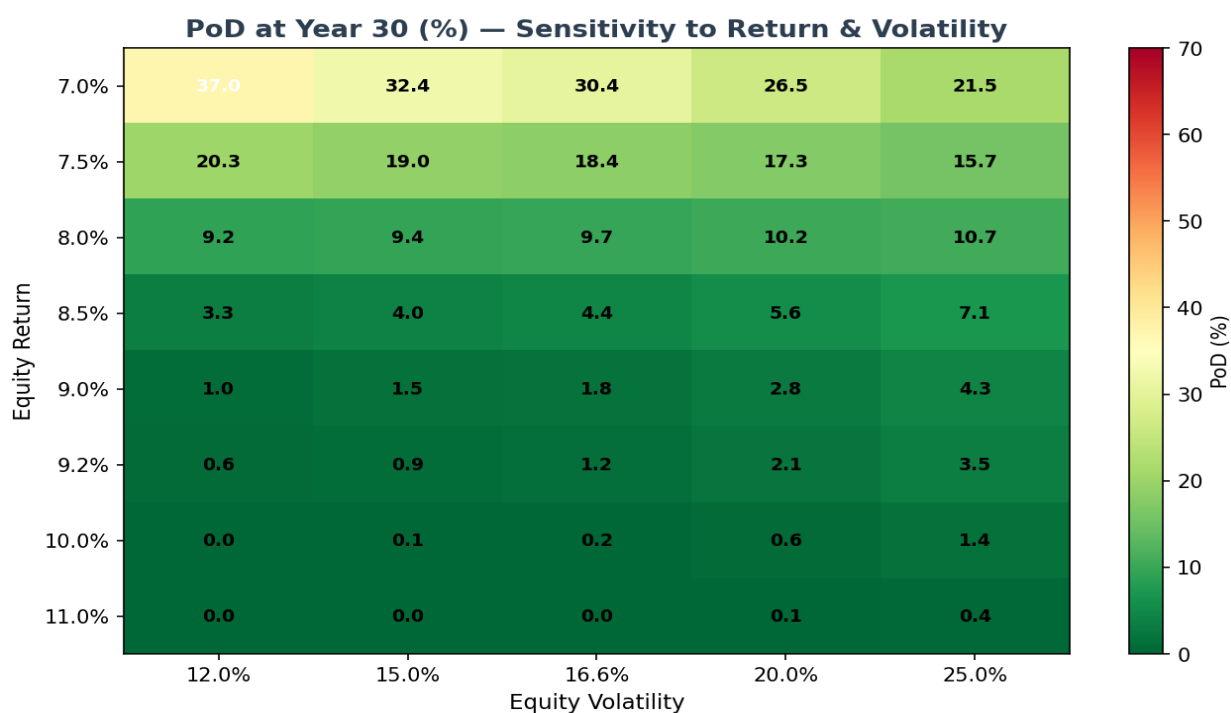
Surplus Distribution — Fan Chart



Metric	Value	Notes
PoD (Year 30)	1.2%	SE: 0.05%
Mean Surplus	\$2,435,992	Average across all paths
Median Surplus	\$2,151,975	50th percentile
1st Percentile	\$-55,115	Worst 1% of paths
5th Percentile	\$508,449	
10th Percentile	\$808,302	
25th Percentile	\$1,361,489	
75th Percentile	\$3,194,847	
90th Percentile	\$4,419,493	
99th Percentile	\$7,476,780	
Conditional Expected Deficit	\$-333,305	Mean loss given deficit
Mean Holiday Years (of 30)	4.5	Average path: 0; mean across all simulated paths
LMI Fair Premium (PV)	\$1,716	Discounted at 2.13%
Tail Risk PoC	0.24%	Beyond top cover limit

Sensitivity Analysis — Equity Return vs Volatility

The sensitivity grid tests 40 combinations of equity return (7.0–11.0%) and volatility (12.0–25.0%). All other parameters held at base case values. Each cell shows PoD at year 30 from a 50,000-path simulation.



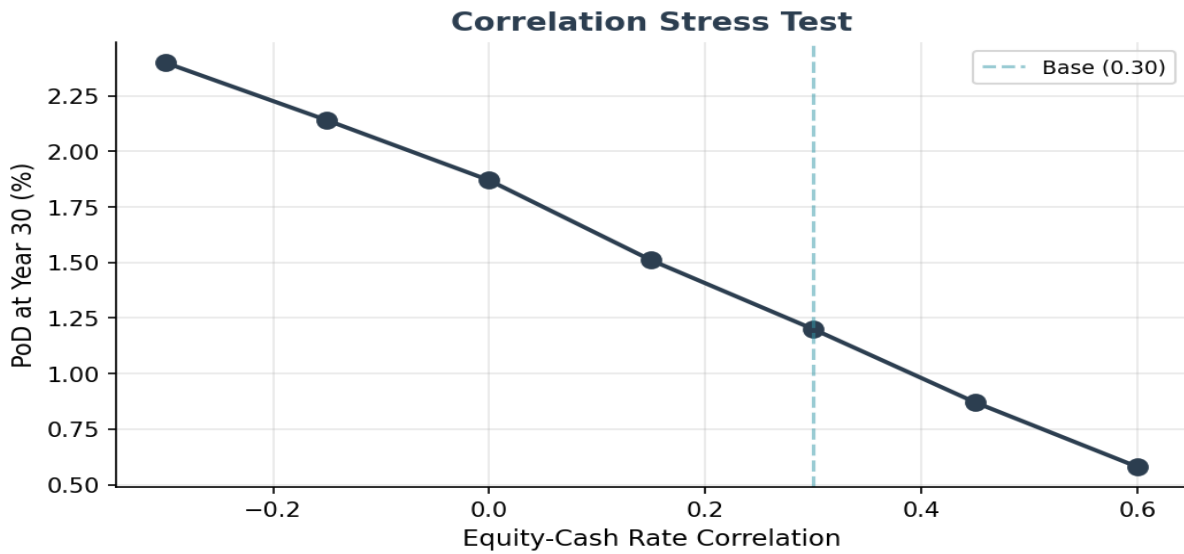
PoD (%) at Year 30 — Equity Return vs Volatility

Return \ Vol	12.0%	15.0%	16.6%	20.0%	25.0%
7.0%	37.0	32.4	30.4	26.5	21.5
7.5%	20.3	19.0	18.4	17.3	15.7
8.0%	9.2	9.4	9.7	10.2	10.7
8.5%	3.3	4.0	4.4	5.6	7.1
9.0%	1.0	1.5	1.8	2.8	4.3
9.2%	0.6	0.9	1.2	2.1	3.5
10.0%	0.0	0.1	0.2	0.6	1.4
11.0%	0.0	0.0	0.0	0.1	0.4

Key observations:

Every 1pp of equity return is worth roughly 3–5pp of PoD. Every 5pp of volatility adds 5–10pp. The base case (9.2% / 16.6%) sits comfortably at 1.2%. The mean-reverting model is central — it reduces long-term tail risk by pulling returns back toward trend after deviations.

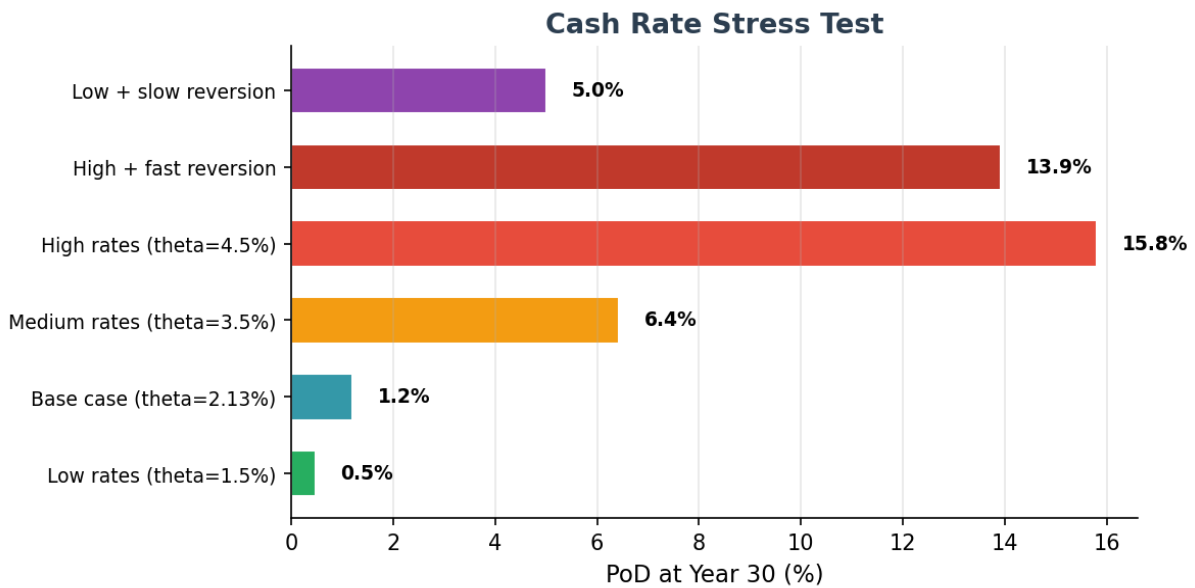
Correlation Stress Test



Correlation	PoD	Mean Surplus	Cond. Deficit
-0.30	2.4%	\$2,419,794	\$-456,206
-0.15	2.1%	\$2,423,903	\$-427,214
0.00	1.9%	\$2,427,041	\$-390,684
0.15	1.5%	\$2,431,662	\$-357,464
0.30 (base)	1.2%	\$2,435,992	\$-333,305
0.45	0.9%	\$2,440,356	\$-310,871
0.60	0.6%	\$2,445,275	\$-267,802

Interpretation: Correlation has minimal impact — the full range from -0.3 to +0.6 moves PoD by only ~2.3pp. This is because the cash rate mean-reverts quickly ($\kappa=0.24$) and the funding cost is dominated by the wholesale margin (2%) rather than the cash rate level. **Correlation mis-specification is not a material risk for this product.**

Cash Rate Stress Test



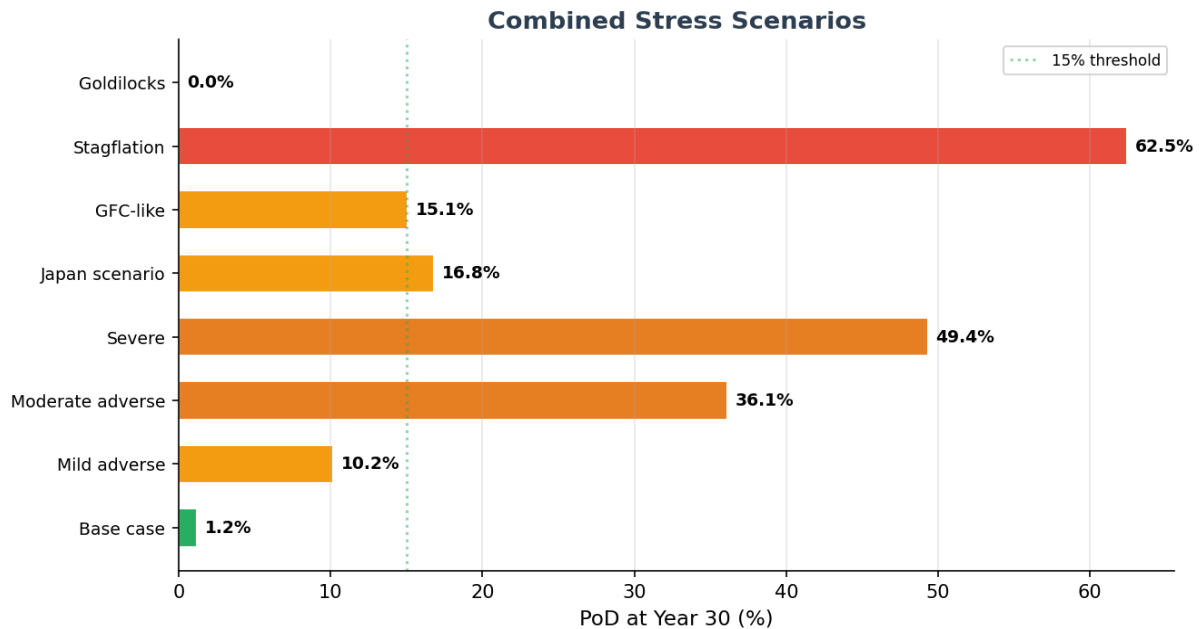
Scenario	Theta	Kappa	PoD	Mean Surplus
Low rates (theta=1.5%)	1.50%	0.24	0.5%	\$2,800,688
Base case (theta=2.13%)	2.13%	0.24	1.2%	\$2,435,992
Medium rates (theta=3.5%)	3.50%	0.24	6.4%	\$1,634,200
High rates (theta=4.5%)	4.50%	0.24	15.8%	\$1,087,025
High + fast reversion	4.50%	0.5	13.9%	\$1,050,697
Low + slow reversion	1.50%	0.1	5.0%	\$2,194,523

Interpretation: Cash rates matter significantly — the spread between low (1.5%) and high (4.5%) theta is 15pp of PoD. Higher cash rates directly increase wholesale funding costs (2.0% + cash rate), accelerating the cost drag. The base assumption (theta=2.13%) reflects a low-rate environment. If the neutral rate settles at 3.5% (plausible post-COVID), PoD rises to 6.42%. At 4.5%, PoD rises to 15.81%.

Note: faster mean reversion (kappa=0.5) barely changes the high-rate result — what matters is where rates settle, not how fast they get there. However, slow reversion (kappa=0.1) at low theta increases PoD to 5.0% because the cash rate takes longer to fall from its current 4.21% starting point.

Combined Stress Scenarios

Combined scenarios test multiple adverse conditions simultaneously, representing plausible macroeconomic regimes over a 30-year horizon.



Scenario	PoD	Median Surplus	Cond. Deficit	LMI Fair
Base case	1.2%	\$2,151,975	\$-333,305	\$1,716
Mild adverse	10.2%	\$1,213,437	\$-476,020	\$21,096
Moderate adverse	36.1%	\$366,723	\$-725,191	\$76,789
Severe	49.4%	\$25,144	\$-1,057,764	\$114,788
Japan scenario	16.8%	\$537,872	\$-327,824	\$34,543
GFC-like	15.1%	\$1,387,860	\$-717,772	\$41,219
Stagflation	62.5%	\$-366,432	\$-1,064,094	\$128,471
Goldilocks	0.0%	\$4,626,009	\$-359,278	\$20

Key observations:

- The **Goldilocks scenario** (11% return, 12% vol, moderate rates) produces PoD of just 0.02% — essentially zero risk.
- The **Japan scenario** (low returns but low vol and low rates) gives 16.83% PoD — low rates partially offset low returns.
- **Mild adverse** (8% return, 20% vol) pushes PoD to 10.19% — a material deterioration from base case.
- **Stagflation** (low returns + high vol + high rates) is 62.45% PoD. This is the scenario the product must insure against.
- The **GFC-like** scenario (25% vol, negative correlation) gives 15.09% — sustained 2008-level volatility for 30 years is extremely adverse.

Actuarial Assessment

Is the Model Sound?

The model structure is sound. The v14c (003) model is well-constructed: the Vasicek/OU process for cash rates uses exact discretisation with MLE-estimated parameters; the GBM+MeanRev equity model is grounded in Shevchenko (2026); the insurance structure (LMI + tail risk reinsurance) is correctly implemented; and the Payments Waterfall provides genuine portfolio-level risk mitigation.

The **critical qualifier** is that headline results are highly sensitive to three specific input parameters — equity drift μ , mean-reversion strength γ , and collar cost — each of which has a wide empirical confidence interval relative to the base-case point estimate. The Model Assumptions & Parameter Risk companion paper performs an independent calibration on S&P 500 TR 1988–2024 data. Its findings, which qualify the base-case PoD reported in this paper, are:

Parameter	Model	Empirical	Impact on PoD
Mean-reversion γ	0.163	MLE 0.173; 95% CI [0.019, 0.447]	$\gamma = 0.10 \rightarrow$ PoD 3.1%; $\gamma = 0 \rightarrow$ 31.5%
Equity drift μ	9.2%	MLE 11.0%; CAPE-implied forward 5–6%	$\mu = 8.0\% \rightarrow$ PoD 9.7%; $\mu = 7.0\% \rightarrow$ 30.4%
Collar cost	0.046% p.a.	Black-Scholes central 0.30–0.40%	0.30% $\rightarrow$ PoD 2.0%; 1.0% $\rightarrow$ 7.6%

Are the Numbers Investable?

At base-case parameters, yes. A 1.2% PoD with a conditional deficit of \$333,305 against a \$2M property is a reasonable risk profile. The LMI fair premium of \$1,716 is commercially viable. The portfolio-level PoC after the Payments Waterfall is well below the 1% threshold that reinsurers typically require under the model as written.

Under an empirically-centred parameter set, the product remains viable but is materially less favourable. Joint re-simulation at $\gamma = 0.13$, $\mu = 8.5\%$, $\theta = 2.7\%$, collar = 0.30%, margin = 2.2% produces PoD = 16.9% and mean surplus = \$995K. Under an adverse-plausible set ($\gamma = 0.10$, $\mu = 8.0\%$, $\theta = 3.0\%$, collar = 0.40%, margin = 2.5% — each parameter inside its empirical range, not at the tail), PoD = 43.2% and mean surplus = \$253K.

This is the dominant finding of the review: parameter uncertainty alone — before any model-structure uncertainty (Gaussian shocks, no jumps, no regime switches) — produces a PoD range spanning more than an order of magnitude. Capital allocation, reinsurance attachment, and investor risk disclosure should be anchored on the adverse-plausible leg rather than the base case.

Recommendations

1. Report headline PoD as a range, not a point. The model-as-written produces 1.2% PoD. Realistic-central (empirically-centred parameters) produces 16.9%. Adverse-plausible produces 43.2%. All three should appear alongside each other in internal, investor, and reinsurer materials. The base case alone overstates confidence in the product.

2. Anchor capital and reinsurance sizing on the adverse-plausible leg. Reinsurance attachment, wholesale-funder subordination, and internal solvency capital should be calibrated to the 43% PoD scenario, not the 1.2% base case. Designing to the base case embeds a material under-reserve.

3. Use conservative μ for pricing. At $\mu = 8.0\text{--}8.5\%$ (rather than 9.2%), PoD sits at 4–10% — still competitive as a product — without reliance on in-sample 1988–2024 US equity performance continuing for the next 30 years. Forward-looking estimates that account for current valuation (CAPE-implied 5–6%) are lower; 8.5% is a defensible middle ground.

4. Replace the collar cost assumption. The 0.046% model input is inconsistent with Black-Scholes pricing at current AU rates (0.16–0.44%), before skew or transaction costs. Move to 0.30% minimum for pricing. Re-test at 0.50% to stress for skew and roll slippage.

5. Governance: annual re-estimation of γ . The mean-reversion parameter is the single most load-bearing modelling choice. Establish a policy for annual MLE refresh with each year of added data, track drift against the 0.163 assumption, and hold reserves against downward revision.

6. Track θ to AU cash, not US Fed Funds. The 2.13% long-run mean is 50bp below the US MLE and 250bp below the AU analogue. Update the cash process to match the funding currency. At $\theta = 3.0\%$ (still below the AU historical mean), PoD is 6.4%.

7. Stress test the wholesale margin annually. The 2.0% modelled margin is at the low end of the 2.0–3.0% range priced for comparable AU wholesale risk. Lock terms for the warehouse facility at issuance; re-test economics at 2.5% on each new tranche.

8. Increase simulation paths in the spreadsheet tool to 10,000+. The production spreadsheet runs 1,000 paths and produces PoD estimates with unacceptable standard error at the base-case rate. 10,000 paths is the minimum for internal decisioning.

9. Document the Shevchenko model and its parameter uncertainty thoroughly. For investor and reinsurer materials, disclose the γ bootstrap distribution, the LRT against $\gamma = 0$, the small-sample bias finding, and the full range of headline PoD across the joint parameter scenarios.

EPM vs Current Investment Landscape

To place the EPM risk profile in context, we benchmark its key actuarial metrics against established credit instruments and traditional residential lending. The table below presents the EPM under both the base-case parameters (as modelled) and the realistic-central parameter scenario developed in the companion Model Assumptions paper. The adverse-plausible scenario is discussed separately in the commentary.

Metric	EPM (base)	EPM (realistic)	Prime AU Mortgage	Moody's AA Bond	AU RMBS (AA)
Cumulative PoD (30yr)	1.2%	16.9%	3–5%	~1.5%	~1–2%
Loss severity (LGD)	~21% of peak	~21% of peak	20–40%	40–60%	Subord.
Expected loss (30yr)	0.13%	~2.3%	0.6–1.25%	0.6–0.9%	0.05–0.15%
Ratings equivalent	AA/AAA	BBB-ish	A/BBB	AA	AA

Actuarial Assessment of Comparative Risk Position

- Credit quality depends on which scenario is selected.** At base-case parameters, the EPM's 1.2% cumulative PoD places it between Moody's AAA and AA 30-year cumulative default rates. At realistic-central parameters (empirically-centred μ , γ , collar cost), the 16.9% PoD corresponds more closely to investment-grade speculative territory (BBB to BB range on a pure-PoD basis). The low 21% severity moderates the expected loss, but no single rating can be attached to the product without first committing to a parameter set.
- Comparison to traditional residential lending is parameter-dependent.** Standard Australian prime mortgages carry a 3–5% cumulative default rate over 30 years. At base-case parameters the EPM is 3–4× safer on a PoD basis; at realistic-central parameters it is of the same order as prime mortgage risk; at adverse-plausible parameters (43.2% PoD) it is materially riskier than prime mortgages. The structural features (mean reversion, collar, holiday, run-off) remain protective across all scenarios, but the magnitude of protection depends on whether the underlying drift and mean-reversion inputs hold.
- Risk driver is fundamentally different from traditional mortgages.** Traditional mortgage default risk is driven by borrower income, employment, and interest-rate affordability. EPM risk is driven by long-term equity-market performance relative to interest rates. These factors are largely uncorrelated, which preserves a diversification benefit for reinsurers regardless of scenario — this structural feature is robust to the parameter uncertainty discussed above.
- Structural protections are real but not unconditional.** Mean-reverting equity, collar hedging, holiday mechanism, P&I; amortisation, and insurance coverage from \$0 form a genuine multi-layered protection framework. However, the protective value of the collar and of mean reversion is itself sensitive to parameter calibration — a collar priced at 0.30% (Black-Scholes) rather than 0.046% (model input) meaningfully erodes the base-case surplus, and a weaker γ removes the self-correcting feature on which the 1.2% base-case PoD depends.

Conclusion

The v14c (003) model is a structurally sound quantitative framework. The Monte Carlo engine, the Shevchenko mean-reverting equity process, the Ornstein-Uhlenbeck cash-rate model, and the treatment of the collar, holiday, P&I; amortisation and insurance layers are implemented correctly and consistently. As a piece of quantitative engineering, the model is credible.

The headline base-case results (PoD 1.2%, mean surplus \$2,435,992) are, however, contingent on three parameter assumptions that each carry wide empirical confidence intervals: equity drift μ , mean-reversion strength γ , and collar cost. The companion paper "Model Assumptions and Parameter Risk" (April 2025) shows that moving each of these to an empirically-centred estimate produces a joint PoD of 16.9%, and that moving to the lower bound of each confidence interval produces a joint PoD of 43.2%. The headline result should therefore be reported as a range, with explicit disclosure of the scenarios that produce each point.

Under realistic-central parameters, the product remains viable and insurable, but at a materially higher claims rate than the base case suggests. Capital and reinsurance sizing should be calibrated to the adverse-plausible leg, not the base case. Pricing should use conservative μ (8.0–8.5%) and a Black-Scholes-anchored collar cost (0.30% minimum). With these adjustments, the product is defensible to an external actuary, investor, or reinsurer. Without them, the 1.2% headline overstates the confidence the parameter evidence supports.

Appendix: Methodology

Component	Model	Parameters
Equity Returns	GBM + Mean Reversion (Shevchenko 2026)	$\mu=9.2\%$, $\sigma=16.6\%$, $\gamma=0.163$
Volatility Buffer	Cap/Floor on annual returns	Cap: +40%, Floor: -20%
Cash Rate	Ornstein-Uhlenbeck (exact discretisation)	$\theta=2.13\%$, $\kappa=0.24$, $\sigma=1.22\%$, $r=4.21\%$
Equity-Rate Correlation	Cholesky decomposition	$\rho = 0.30$
Collar Hedge	Calculated put-call	Net cost: +0.046% p.a. (near zero)
Variable Costs	Fixed spread	3.45% (wholesale 2.0% + NIM 0.70% + hedge 0.25% + FP 0.50%)
P&I; Amortisation	Linear paydown Years 11-30	Peak \$1.6M, annual \$80K repayment
Interest Payment Holiday	Entry 0.75×, Exit 1.458×	Interest payments pause; annuity to borrower continues
Insurance Discount	Continuous ($\exp(-\theta T)$)	Rate = 2.13% (cash rate theta)
LMI Coverage	80% of deficit paths	Top cover limit: P20 of deficit distribution
Tail Risk	20% of deficit paths	Excess beyond top cover limit
Profit Share	10% every 5 years	On positive surplus only

Component	Model	Parameters
Simulation	50,000 paths	Seed: 42, NumPy default_rng